

TCP CONNECTION PERFORMANCE ANALYSIS

Using Mininet and Wireshark

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1. Mininet Topology

The experiment was performed using a simple Mininet TCP client-server topology. Two hosts, s1 and s2, were connected through a virtual network and used to exchange TCP messages. The server listened on TCP port 5000, while the client established a connection and sent multiple messages. The topology allowed the performance of persistent and new-connection modes to be compared under the same network conditions.

- Client host: 10.0.0.2
- Server host: 10.0.0.1
- Server TCP port: 5000
- Transport protocol: TCP
- Traffic was captured using Wireshark for verification.

2. TCP Client–Server Communication

TCP provides reliable, connection-oriented communication between a client and a server. Before application data is exchanged, the client and server establish a connection using the TCP three-way handshake. The client sends a SYN packet, the server replies with SYN-ACK, and the client completes the process with an ACK. After this handshake, application data can be transferred reliably.

In persistent mode, the same TCP connection is reused for multiple messages. Therefore, the connection setup cost is paid only once. In new-connection mode, a separate TCP connection is created for each message. Every message therefore requires a new handshake and additional connection-management overhead.

3. Persistent and New-Connection Modes

Feature	Persistent	New Connection
TCP connection	One connection reused	New connection for each message
Handshake overhead	Only at the beginning	Repeated for every message
Response time	Lower	Higher
Efficiency	Better for many messages	More overhead
Best use case	Continuous communication	Independent short transactions

4. Experimental Results

The measured results show a clear performance difference between the two connection modes. For 50 messages, persistent mode produced an average response time of approximately 210 ms, while new-connection mode produced approximately 405 ms. Thus, persistent TCP communication was significantly faster in this experiment.

Mode	Bandwidth (Mbps)	Delay (ms)	Message Size (Bytes)	Avg. Response Time (ms)	Throughput (msg/s)
Persistent	5	50	128	210.3	4973.64
Persistent	5	50	512	210.3	4951.64

Assignment 2 – TCP Connection Performance Analysis

Persistent	5	50	1024	203.1	9743.10
New Connection	5	50	128	203.1	4630.23
New Connection	5	50	512	205.4	4927.70
New Connection	5	50	1024	208.2	4918.35

Experimental Result Table Screenshot

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roll_no,name,mode,bandwidth_mbps,delay_ms,message_size_bytes,total_messages,average_response_time_seconds,throughput_bytes_per_second,status
CS-BTC25-39,PARINITA DEVI,persistent,5,50,128,10,0.1012,1264.82,Success
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CS-BTC25-39,PARINITA DEVI,new_connection,5,50,128,10,0.2031,630.23,Success
CS-BTC25-39,PARINITA DEVI,new_connection,5,50,512,10,0.2054,2492.7,Success
CS-BTC25-39,PARINITA DEVI,new_connection,5,50,1024,10,0.2082,4918.35,Success
```

Figure 5. Experimental output showing the measured TCP performance results.

5. Graphical Analysis

The following graphs summarize the measured TCP performance. The graphs compare average response time between persistent and new-connection modes, show the effect of message size, and show response time over a sequence of messages.

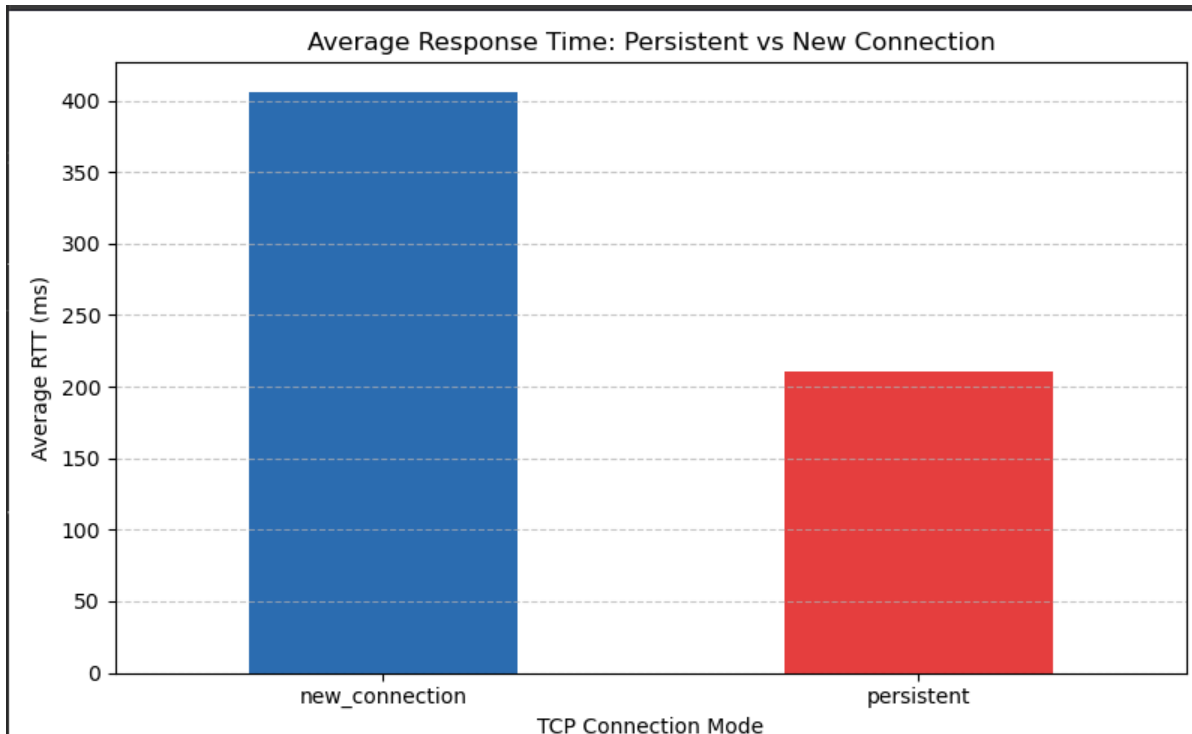


Figure 1. Average Response Time: Persistent vs New Connection (Image 3).

This graph shows that the new-connection mode has a much higher average response time than the persistent mode. The difference is mainly caused by repeated TCP connection establishment.

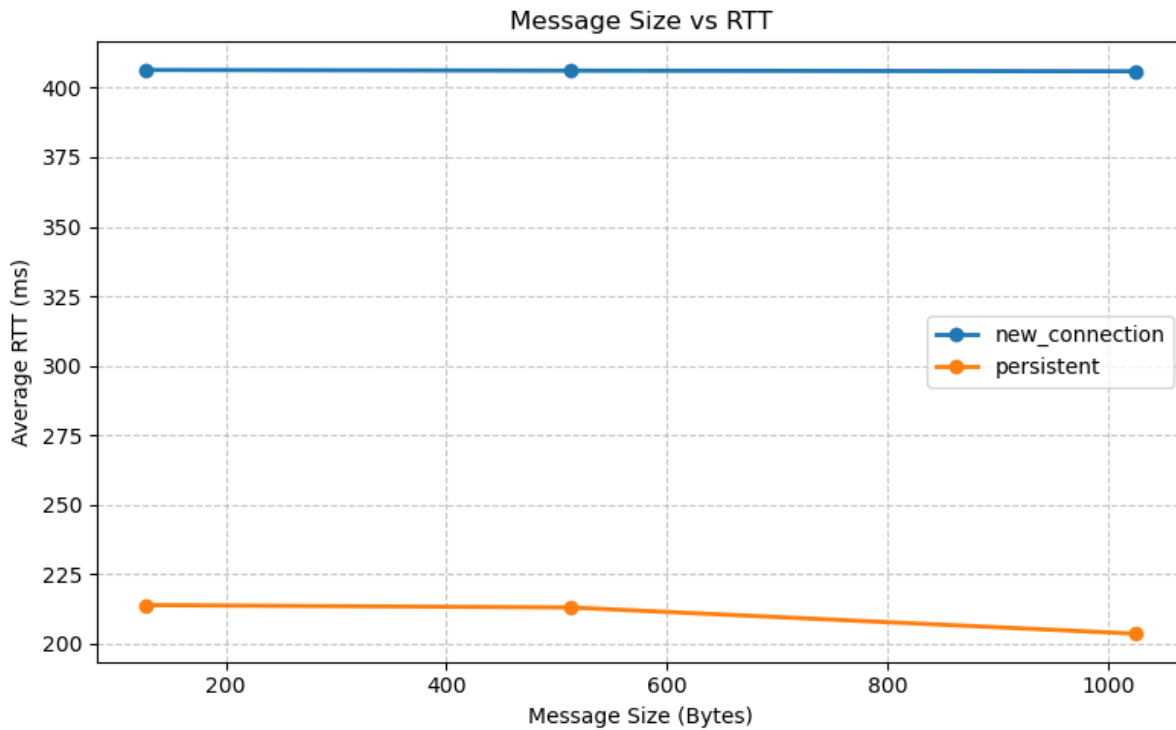


Figure 2. Effect of Message Size on RTT (Image 4).

The graph indicates that message size has a relatively small effect on response time in the measured range. Persistent mode remains consistently faster, while new-connection mode remains close to 400 ms because the connection setup overhead dominates.

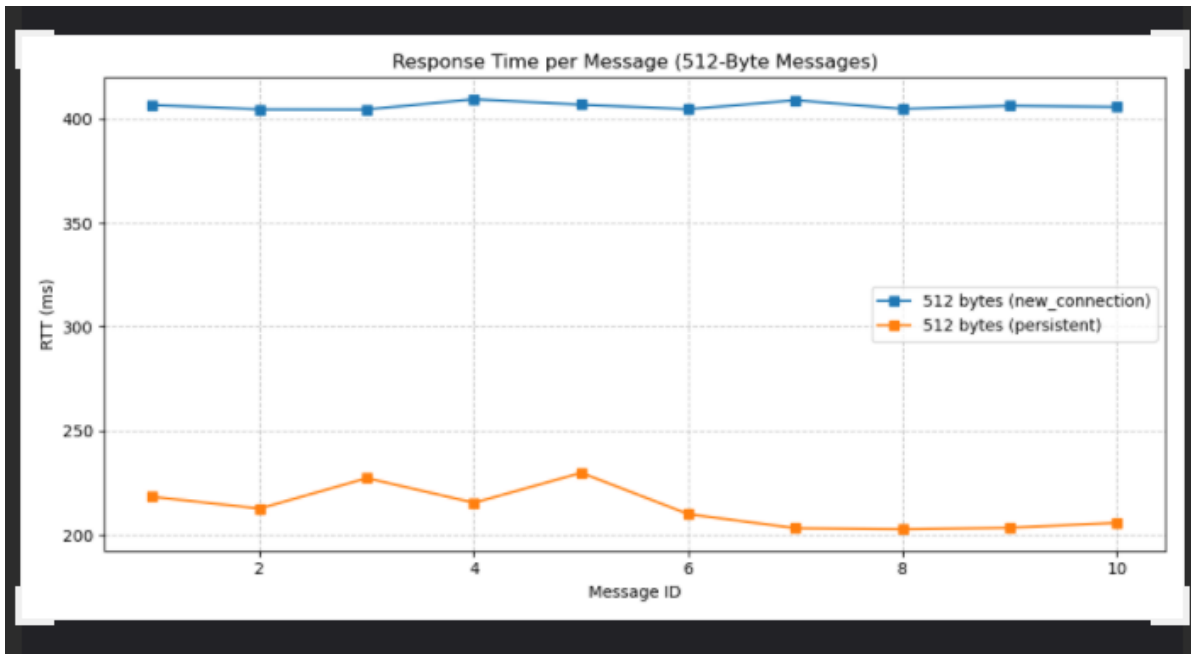


Figure 3. Response Time per Message for 512-Byte Messages (Image 5).

The per-message plot demonstrates that persistent communication remains around 200–230 ms, whereas new-connection communication remains around 400–410 ms. This confirms the consistent performance advantage of connection reuse.

6. Wireshark Verification

Wireshark was used to capture and inspect the TCP packets generated during the experiment. The capture shows TCP traffic between 10.0.0.2 and 10.0.0.1 on port 5000. The packet list includes SYN, SYN-ACK, ACK, and PSH/ACK packets, which confirms both TCP connection establishment and subsequent application-data transfer.

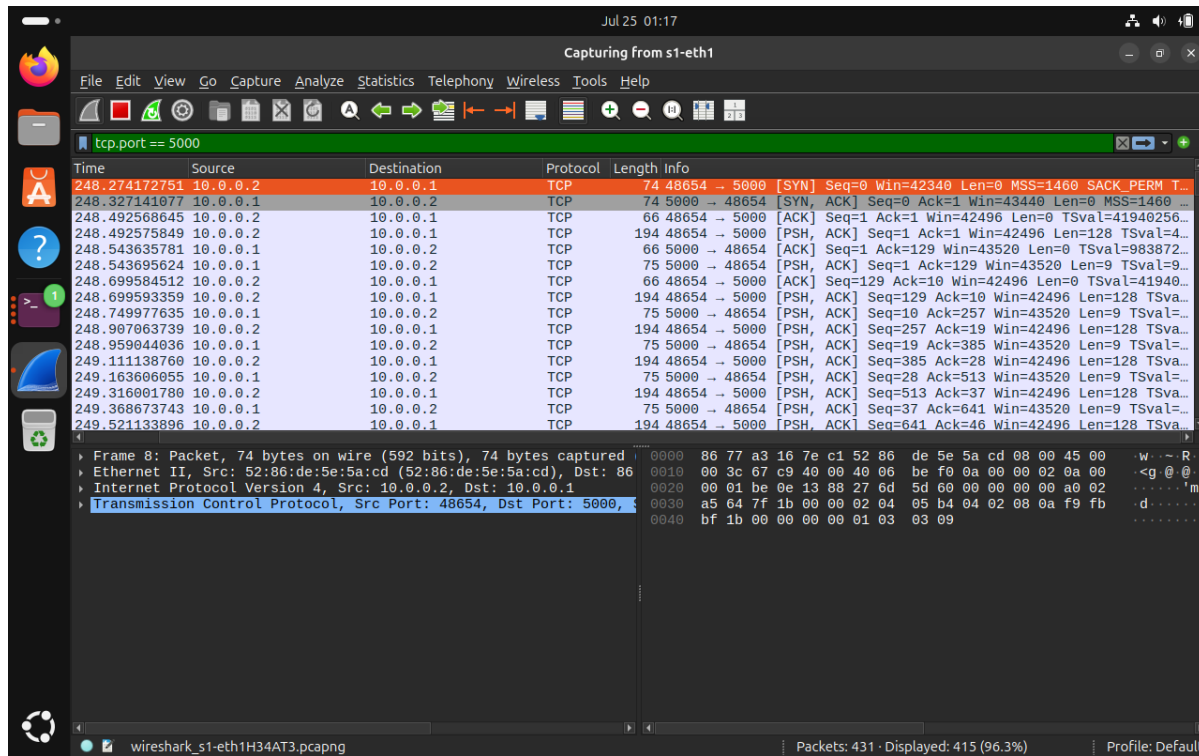


Figure 4. Wireshark TCP Capture Showing the Three-Way Handshake and Data Transfer (Image 2).

The SYN packet from the client to the server, followed by SYN-ACK from the server and ACK from the client, verifies the three-way handshake. The later PSH/ACK packets indicate application data being carried over the established TCP connection. This packet-level evidence supports the performance results obtained from the Mininet experiment.

7. Answers to Questions

1. Why does TCP use a three-way handshake?

TCP uses a three-way handshake to establish a reliable connection and synchronize the initial sequence numbers of both endpoints. The exchange of SYN, SYN-ACK, and ACK confirms that both the client and server are reachable and ready to communicate.

2. Which mode had lower response time?

Persistent mode had the lower response time. The measured average was approximately 210 ms, compared with approximately 405 ms for new-connection mode.

3. Why does new-connection mode have more overhead?

New-connection mode creates a separate TCP connection for every message. Each connection requires a handshake and connection-management operations, which add extra delay and processing overhead.

4. Which mode is better for CampusChat and why?

Persistent mode is better for CampusChat because chat applications exchange many messages between the same users. Reusing an established TCP connection avoids repeated handshakes and reduces response time.

5. Which mode is suitable for NetAttend and why?

New-connection mode can be suitable for NetAttend when requests are short and independent, such as occasional attendance submissions. However, persistent connections may be better if the system sends frequent requests or maintains continuous communication.

6. How does message size affect throughput?

Larger messages can improve throughput because more data is transferred per connection and the fixed overhead becomes relatively smaller. In the experiment, the 1024-byte message case showed higher measured throughput for persistent communication.

7. How did Wireshark help verify TCP behavior?

Wireshark provided direct packet-level evidence of TCP behavior. It showed the SYN, SYN-ACK, and ACK handshake packets and the subsequent PSH/ACK data packets, confirming connection establishment and reliable data transfer.

8. Conclusion

The experiment demonstrates the performance advantage of persistent TCP connections when multiple messages are exchanged between the same client and server. Persistent mode avoids repeated three-way handshakes and therefore achieves lower response time and better efficiency. New-connection mode is simpler for independent transactions but introduces significant connection-establishment overhead. The Mininet results and Wireshark packet capture together confirm the relationship between TCP connection management and application-level performance.